

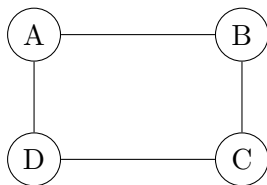
GRAPH THEORY

Activity 2 – Problem 2: Bipartite Graphs and Cycles

Statement. If a graph is bipartite, then it does not contain an odd cycle. Conversely, if a graph contains an odd cycle, then it is not bipartite.

1. Example of a bipartite graph with no odd cycle

Consider the cycle graph C_4 :



The vertices can be separated into two sets,

$$V_1 = \{A, C\}, \quad V_2 = \{B, D\}.$$

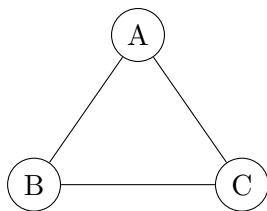
Every edge joins a vertex in V_1 to a vertex in V_2 , so C_4 is bipartite. Its cycle is

$$A - B - C - D - A,$$

which has length 4. Since 4 is even, this graph has no odd cycle.

2. Example of a graph with an odd cycle

Consider the cycle graph C_3 :



The cycle

$$A - B - C - A$$

has length 3, so it is an odd cycle. If A is placed in one part of a bipartition, then both B and C must be placed in the other part because each is adjacent to A . However, B and C are also adjacent to each other. This would place the edge BC inside the same part, which is not allowed in a bipartite graph. Therefore, C_3 is not bipartite.

These two examples illustrate the result: a bipartite graph has no odd cycle, while the presence of an odd cycle prevents a graph from being bipartite.